CLEANQUEST · JUDGES' PRESENTATION

CleanQuest — Judges' Presentation

Meta Global AI Developer Hackathon · Indie track · **Build for Your World**

CleanQuest makes the invisible 2 a.m. cleaning shift visible — night cleaners prove their work with two photos, own a verified record that travels with them job to job, and get recognized instead of re-checked.

Every screenshot below was captured 2026-10-31 from the real app. Meta stack: **Muse Spark 1.3** (per-surface scorecards) + **SAM 3.1** (surface overlays) + **Muse voice transcription** + **Baseten Spark Open** as a logged second provider — plus an on-device MediaPipe + classical-vision rung that plays with no keys and no signal.

▶ Launch Live App | Spectator Board | Download PDF

Indie track: creativity · technical ambition · whether it works

Vanilla JS · no build · no backend · offline-first

EN / ES / PL throughout

Demo-ready: one-tap Judge Demo + live loop + spectator board

How to score this — rubric + MVP map

Criterion (weight)	CleanQuest's answer	MVP coverage (Operating Standards)	Where to see it
Impact 30% — who benefits, does it matter, could people use it	Non-technical night cleaners get portable proof of work; small operators get proof without re-checks; associations get training delivered at the moment of work. Consent-first, faces never identified, read-only appeals, top-10-only boards.	Screens 1–4 (language, phone+PIN, consent + read-aloud, home with streak/week/matchup strip), 10 (crew + top-10 + MVP + personal record), 11–15 (pledge-gated supervisor tools, sites+QR, squad create/switch, training gaps, one-tap packs). All 10 company rules honored.	Run in Spanish for 20s; check the home status strip; open supervisor tools → Squads; file a dispute
Meta AI Tooling 25% — beyond a basic integration	Spark scorecards grade each surface with confidence gating (low-confidence marked unverified, never smoothed over); SAM 3.1 overlays ground every claim in pixels; voice notes transcribed; Baseten Spark Open toggle with exportable per-run log; on-device EfficientDet-Lite0 + Pose Landmarker rung.	Screen 8 (weighted 50/25/25 total + three explanation bars + surface map + provenance line); Settings provider toggle + exportable run log.	Results scorecard with provenance line; provider log export; one-tap Judge Demo
Demo 25% — working, holds up live	One-tap Judge Demo runs the real pipeline on sample photos in ~10 seconds; then run the live loop on a real room; spectator board projects the ranking.	Screens 5–7 (registered-site picker + QR check-in + system-assigned spot + frame guide; timer with neutral live grid so misses stay hidden; ghost overlay for after-capture); Screen 9 (5-card trilingual swipe deck + labeled post-answer sponsor slot); spectator board.	Instructions → Judge Demo; camera site picker + QR; `?spectator=1` on the big screen
Technical Depth 20%	11k lines of dependency-free vanilla JS; hermetic	All 9 Experience Principles (under 3 min, zero typing in the	Settings → run self-test; airplane-

— sound, refined engineering	self-test + persona suite; offline queue that re-scores on reconnect; 44px touch floors verified on 9 mobile screens + desktop; no fake data paths (demos excluded from ranking, samples labeled and ineligible).	run, language-first, 44px glove targets, offline queue + rescore, low-end phone downscaling, celebrate-never-shame, explain every score, hype host default-Off with muted tones) + all 6 run-integrity rules (registered-site eligibility enforced).	mode run; this report's appendix

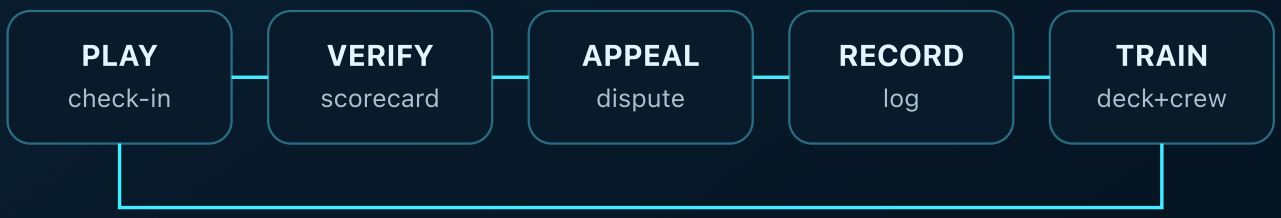
MVP compliance — 15 screens + operating rules

#	MVP screen / rule	Status	Evidence in this build
1–3	Language picker · Phone+PIN · Consent + read-aloud	SUPPORTED	Top-level EN/ES/PL; local hashed 4-digit PIN (no email); first-run consent gate with native MP3/TTS read-aloud, "faces never identified", "never for discipline".
4	Home (Start, matchup, streak, cards, sponsor card, 13-wk meter)	SUPPORTED	Always-visible status strip shows streak + week of 13 + matchup at a glance; full passport/sponsor/meter hub one tap below.
5–7	Check-in (Site + QR + assigned spot + frame guide) · Run timer · Ghost after-capture	SUPPORTED	Registered-site picker + `CQ-QR` scan; daily system-assigned spot; neutral live grid (misses hidden until scored); 4-mode before-shadow ghost overlay.
8	Score reveal (total, 3 bars, surface map, Dispute)	SUPPORTED	50/25/25 weighted hero, Clean/Technique/Time bars, surface map, provenance line, one-tap read-only appeal (5-business-day promise), Judge Demo path.
9–10	Swipe deck (5/day, missed repeat, labeled sponsor) · Crew & leaderboard	SUPPORTED	30 trilingual brand-neutral cards, post-answer `SPONSORED` slot; top-10 + own row only, crew average + system MVP, personal record with opt-in share, empty-state row when no eligible runs exist.
11–15	Admin: Pledge · Sites & QR print · Crew & squads · Training gaps · One-tap packs	SUPPORTED	All panes gated by the non-discipline pledge; site registry feeds check-in + printable V2 QR stickers; squad chips switch active crew + `+ NEW SQUAD` creates 5-seat squads; Where/How/When/Who/Knowledge gaps; one-tap 5-day pack assignment.

Rules	Default-off voice · Misses hidden · Registered-site eligibility · Offline rescore	SUPPORTED	Voice + UI tones respect mute; live grid neutral; `prizeEligible` requires registered site + system spot + real capture; reconnect runs `scoreChallenge` on queued photos.

Not another image analyzer

A one-shot classifier would fail the gate. CleanQuest is a closed loop, and every link is in the build:

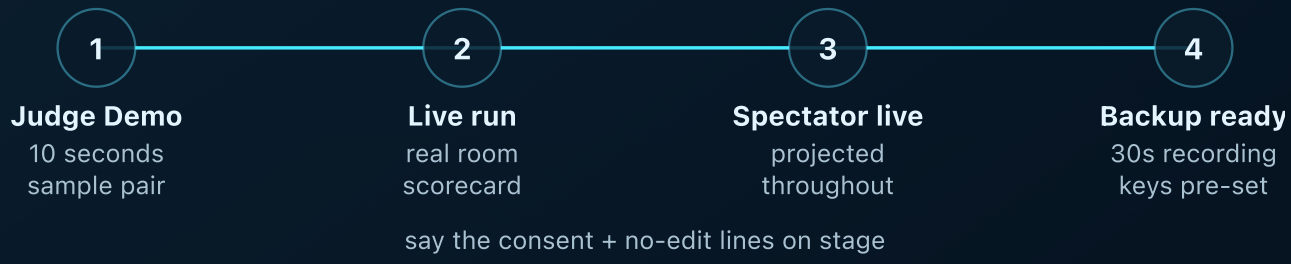


every run feeds the record — nobody can edit a score

- **Play** — check in with a QR, system-assigned high-impact spot, frame guide, in-app capture only.
- **Verify** — three-bar scorecard (Clean 50 / Technique 25 / Time 25), per-surface verdicts with confidence, SAM overlays, provenance line naming the model.
- **Appeal** — one-tap read-only dispute, 5-business-day promise. No person anywhere can edit a score — enforced in code, not policy.
- **Record** — geofenced, exportable inspection log; portable cleaner-owned passport, opt-in employer share.
- **Train & crew** — 30-card trilingual swipe deck with missed-topic repeat, weekly crew matchups, system-chosen MVP, 13-week consistency meter.

Say it on stage: *"The system scores, nobody edits — here is the appeal button that proves it."*

Live demo script (2 runs, ~3 minutes)

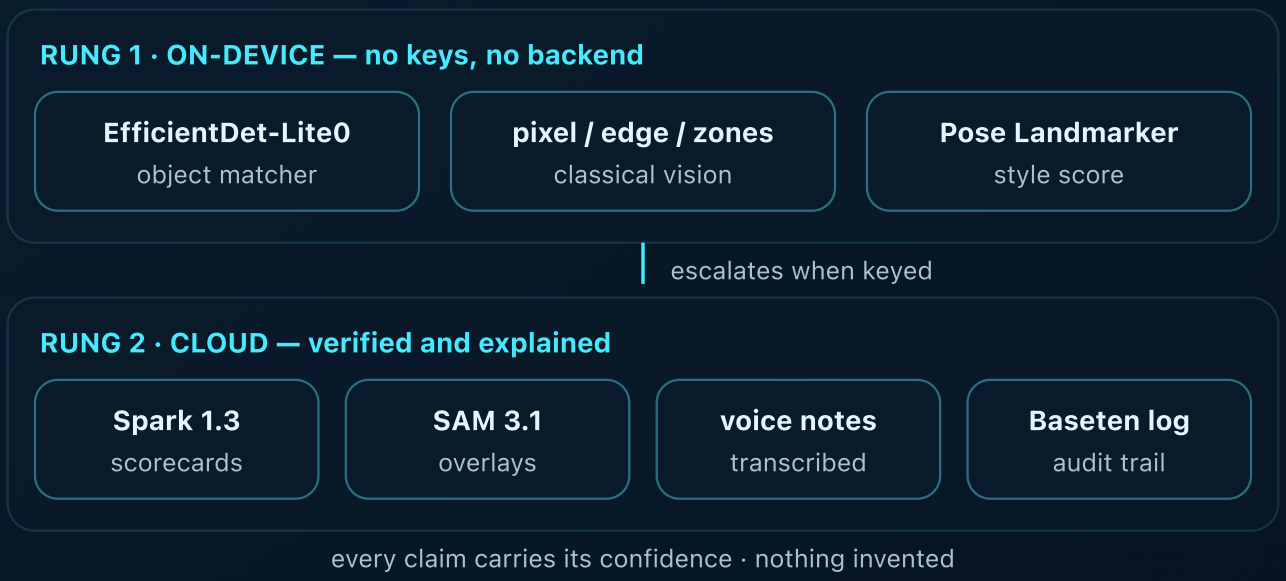


say the consent + no-edit lines on stage

1. **Run 1 — Judge Demo (10 seconds).** Instructions screen → *Judge Demo*. Narrate: *"This is a synthetic dirty/clean pair so you can see the scoring story without us mopping the judging room — same on-device pipeline, watch the provenance line."* Tap through to the scorecard; point at the three bars and the demo label.
2. **Run 2 — live loop.** Check in, shoot a real before/after in the room, finish, show the scored card with its provenance line ("Verified by …" with cloud keys pre-entered, or the on-device line without).
3. **Throughout:** spectator board (`?spectator=1`) projected; host voice explicitly enabled for run 2 (it defaults to Off — say why: occupied buildings). Keep a 30-second backup recording of a full loop in case the room, camera permission, or venue wifi fails.
4. **Before stage:** pre-enter cloud keys in the demo browser, pre-warm MediaPipe on venue wifi (models cache after one load), print area QR stickers, clear localStorage so no prior runs pollute the board.

How the AI works

CleanQuest runs a **two-rung vision ladder** — real on-device perception with no keys and no backend, plus an opt-in cloud rung that verifies and explains. One honest caveat: the classical scoring layer works with zero signal, but the MediaPipe weights fetch from CDN on first use and then cache — a first-ever run on airplane mode scores via pixel-diff plus the neutral pose fallback, with full ML perception on every run after.



Rung 1 — on-device (private by construction, ML after first load)

- **Object counting:** MediaPipe EfficientDet-Lite0 (int8 TFLite, loaded on demand, never bundled) counts objects before/after; a matcher measures "objects cleared". Score threshold 0.35, up to 30 results.
- **Change + structure:** classical vision on a downscaled frame — pixel difference (threshold 38), edge-based structural difference, and a 3×3 zone texture grid that also drives the live zone map.
- **Victory-pose style score:** MediaPipe Pose Landmarker Lite (CPU delegate, single pose) reads 33 landmarks into a 0–10 style score. No model, no person, no camera? It says so plainly and applies a neutral 7.5.
- **Why it matters:** photos never leave the browser on this rung; faces are never identified; no facial recognition or biometrics anywhere.

Rung 2 — cloud verification (opt-in, keyed, explained)

- **Verified scorecards:** `muse-spark-1.3` grades each surface (counter, glass, floor, fixture). Surfaces under 0.6 confidence are marked *unverified*.
- **Surface overlays:** SAM 3.1 segments each surface so the scorecard points at the pixels it means.

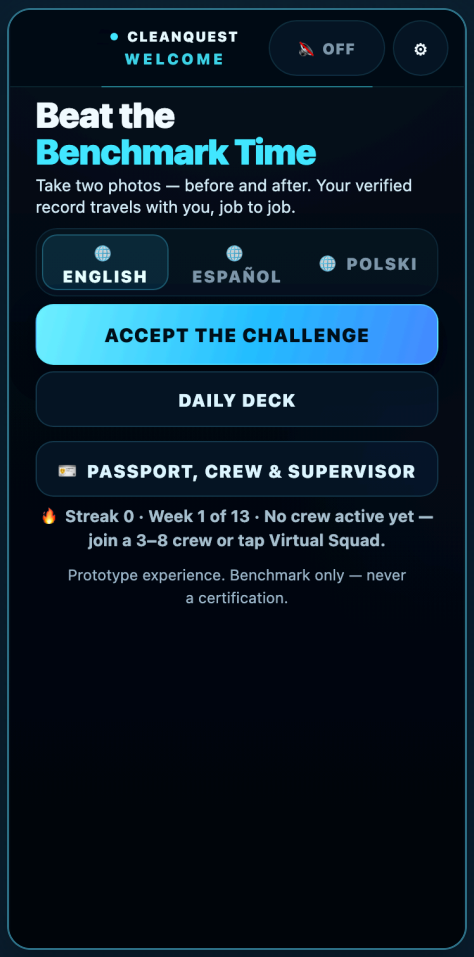
- **Voice notes:** transcribed via Muse voice transcription; without a key the audio attaches untranscribed.
- **Second opinion:** Baseten Spark Open toggle (`meta-models/muse-spark-open-1026`) with an exportable provider log.
- **Graceful failure:** no key or cloud down → the game plays on with an "Offline score" note; queued runs marked "awaiting score" are re-scored with the real pipeline on reconnect; a failed cloud check falls back to the on-device score with the failure stated on screen.

The host voice

231 pre-rendered EN clips (ES/PL via TTS + synth) across verdict tiers routed by margin; scores spoken digit-by-digit so audio always matches the number; one stoppable WebAudio chain; **default Off, never plays uninvited** — including UI tones, which respect the mute.

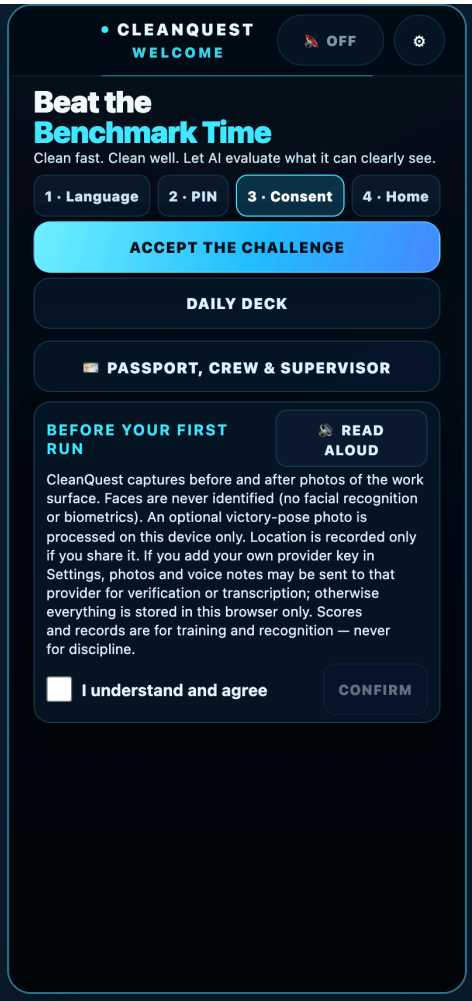
Feature tour — what was built, how it looks

Welcome + onboarding



- **Entry hub:** trilingual picker, Accept the Challenge, Daily Deck, Passport/Crew/Supervisor hub.
- **Always-visible status strip:** streak · week of 13 · crew matchup at a glance.
- **First tap:** plain-language consent with read-aloud, plus optional phone + 4-digit PIN.

READY · MVP 1–4

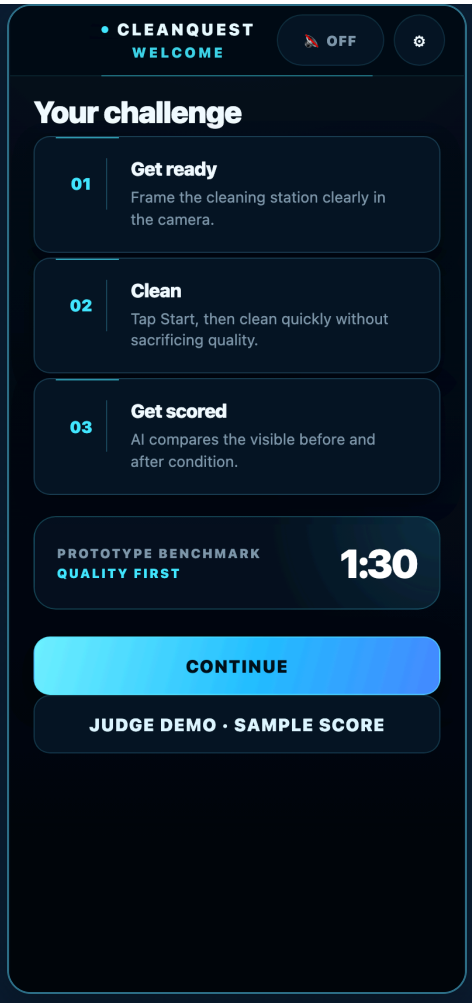


- Consent-open state keeps every CTA ≥44px; CONFIRM visible without scrolling.
- Photos stay on-device; faces never identified; records never used for discipline.



- Full ES translation holds layout — longer strings, no clipping, no truncation.
- The workforce reads first; the app is not translated afterward.

Instructions + one-tap Judge Demo



Camera check-in + timer

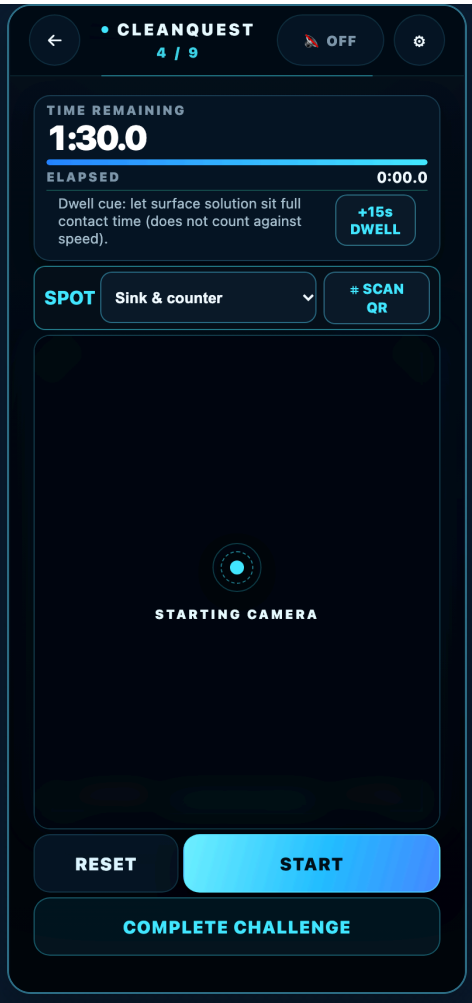
- **Three numbered steps** + benchmark card + consent-gated camera entry.
- **Judge Demo button:** one tap runs the real local pipeline on sample photos and lands on a labeled scorecard — consent-gated exactly like the real path.

READY — the 10-second scoring story



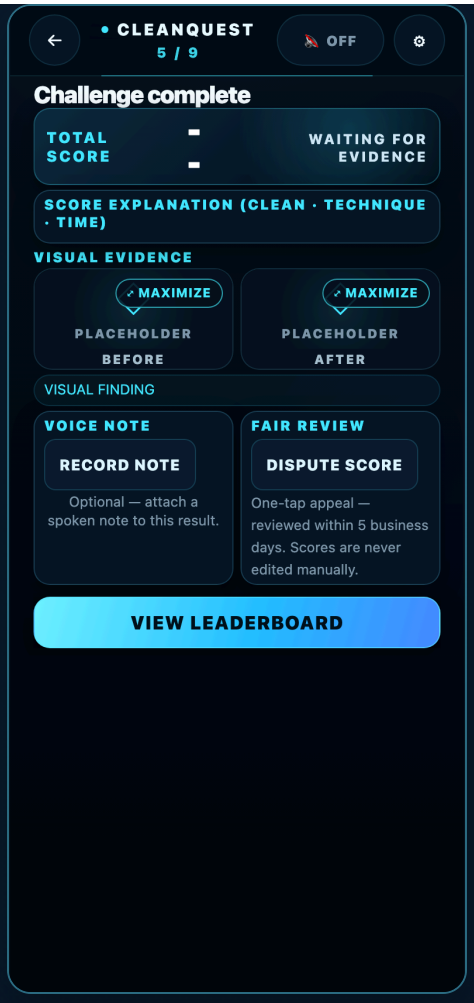
- **Check-in:** registered-site picker + League / Area / system-assigned spot; a selected registered site is required for prize/crew eligibility.
- **Capture:** QR scan, frame guide, in-app capture only.
- **Fit:** short-viewport guard keeps CTAs in the first fold at 360×740.

READY · MVP 5 (needs a real device to play)



- **Run:** countdown, ghost overlay to match the angle, dwell cue, host callouts, big Finish.
- **Integrity:** live zone grid stays neutral during the run — misses remain hidden until scoring, per the operating standards.

Results scorecard



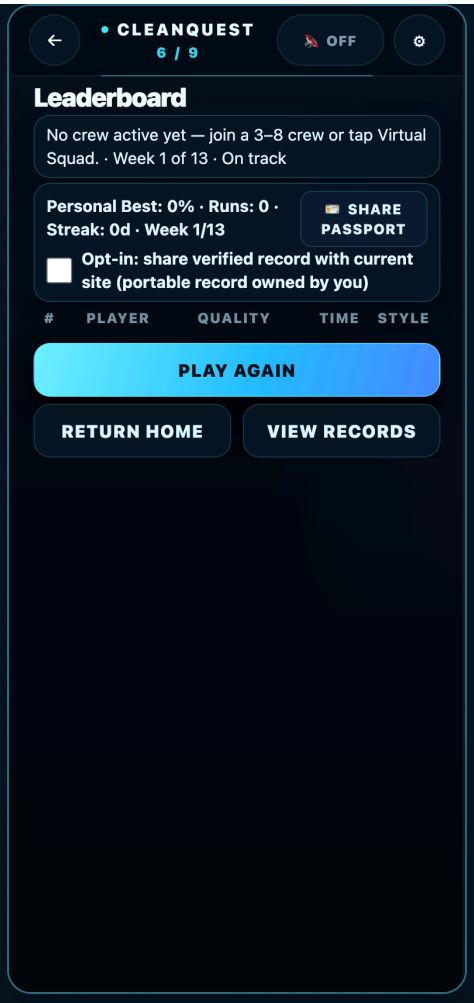
- **Score:** weighted TOTAL hero (50/25/25), three explanation bars, honest surface map.
- **Evidence:** uncropped photos with lightbox; one-tap read-only Dispute; optional voice note; pose score.
- **Provenance line:** provider + model + SAM status, or the on-device line; Judge Demo label on demo runs.

READY · MVP 8



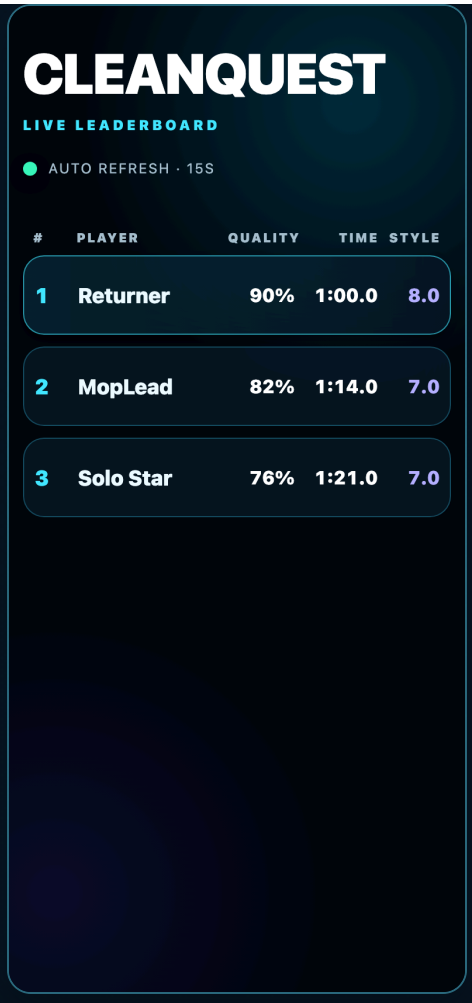
- Real pipeline output on sample photos, **labeled demo**.
- No leaderboard credit, nothing persisted. Headless capture shows the honest model-unavailable fallback; on a device this is a computed score.

Leaderboard + spectator



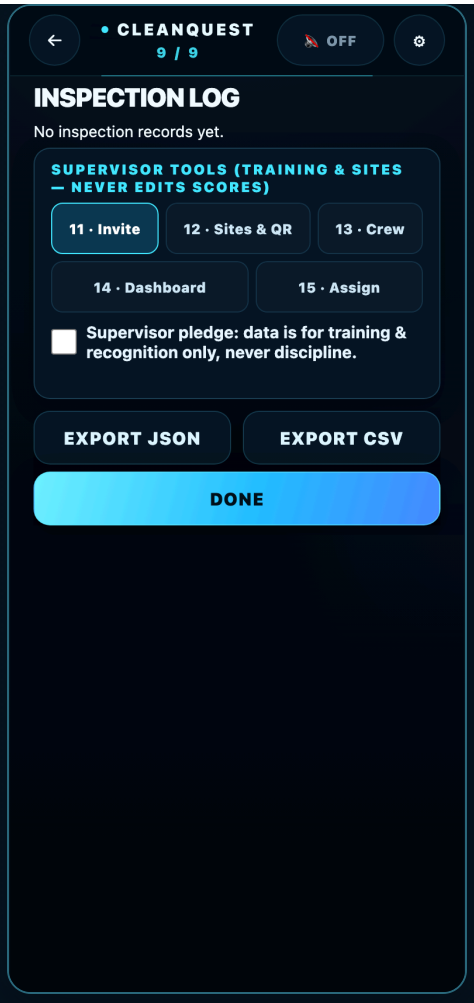
- **Fair board:** top-10 + your own row only; practice and demo runs excluded; labeled sample rows excluded from ranking; personal best computed from eligible runs only.
- **Honest empty state:** fresh installs show an empty board with a labeled empty-state line — clear localStorage before judging.

READY — best demo surface · MVP 10

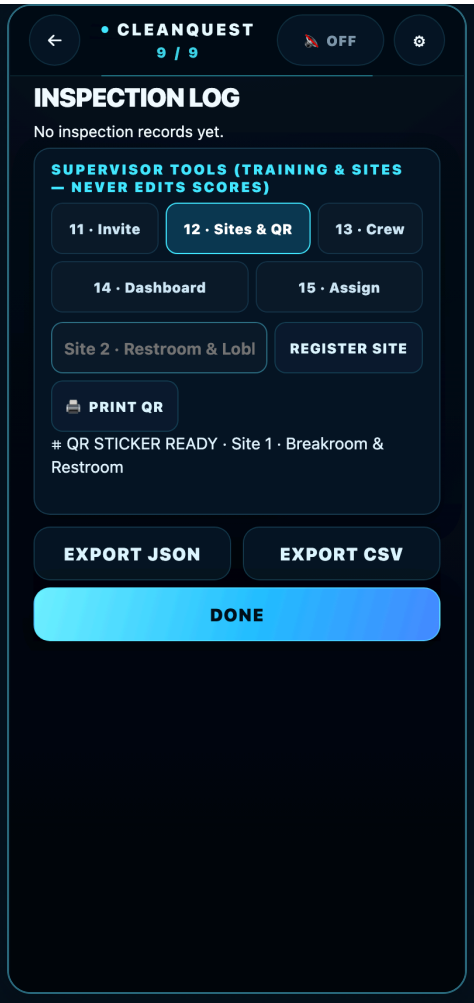


- **Readable at a distance:** rank, quality, time, style — auto-refreshes for projecting.
- Rows shown are manually staged demo entries for illustration; fresh installs start empty until real eligible runs land.

Records, settings, deck



- **Inspection log:** geofenced runs with JSON/CSV export — proof that writes itself.
- **Supervisor panes 11–15:** pledge-gated tools, sites + printable QR stickers, bulk invite, training gaps with one-tap 5-day packs.
- **Offline queue** re-scores on reconnect with the real pipeline.

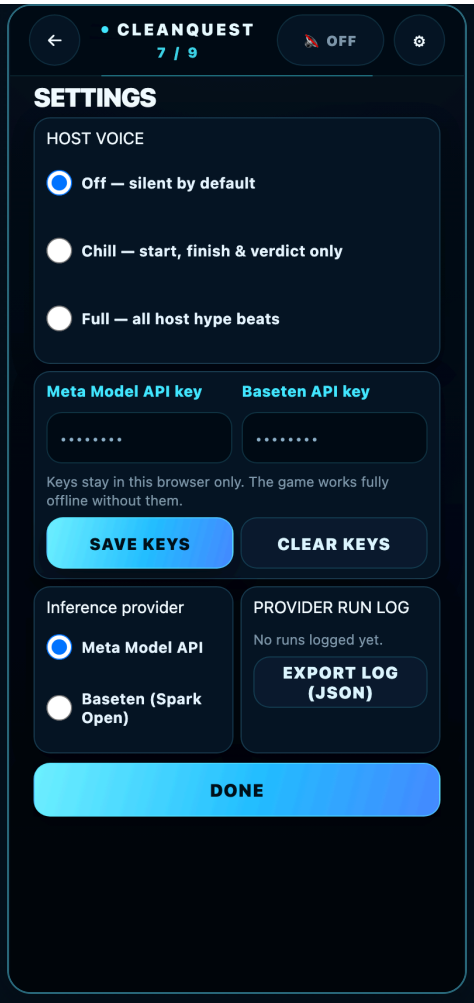


- Sites pane with QR sticker preview — registered sites populate the camera check-in picker and gate prize eligibility.

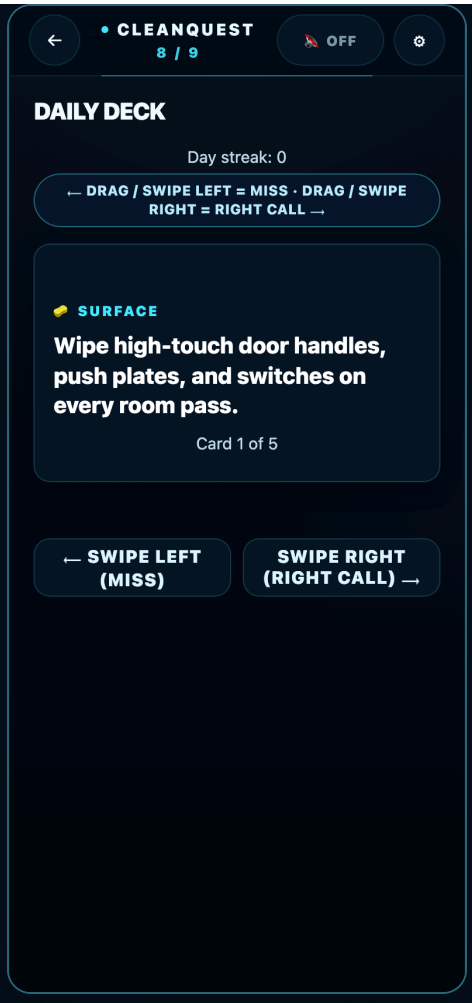


- Squads pane: tap any squad chip to switch the active crew; invite/approve in bulk.
- **+ NEW SQUAD** forms a new 5-seat squad — all 44px targets.

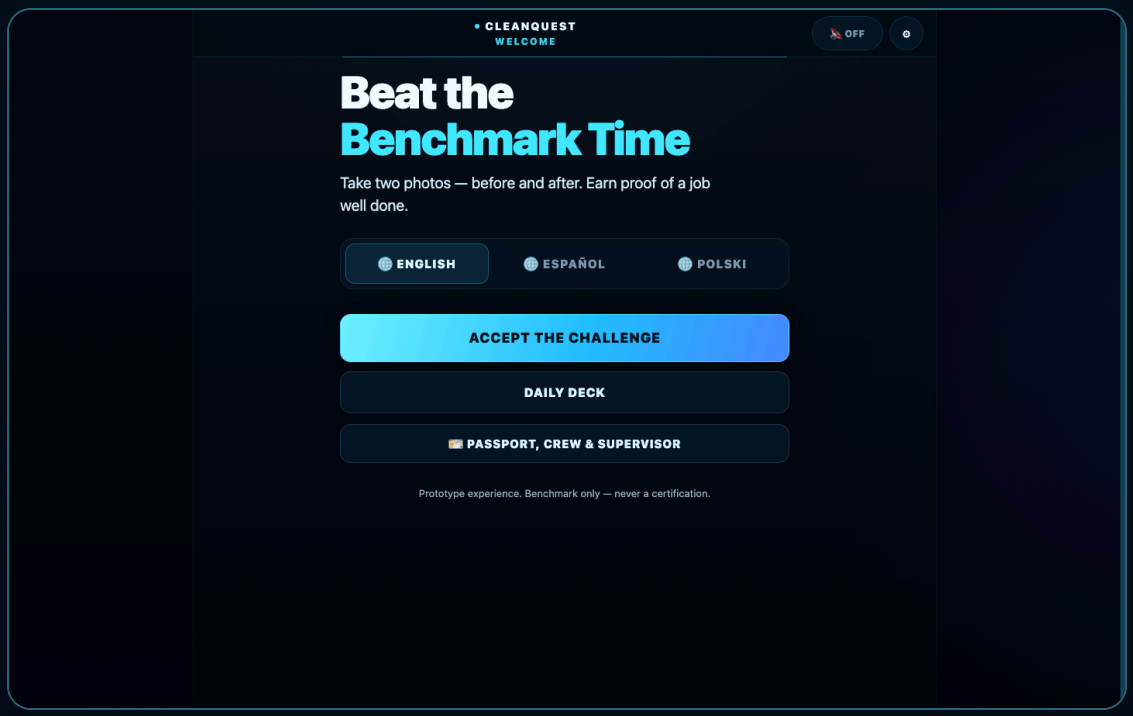
READY · MVP 13



- **Host voice:** default Off, never plays uninvited — including UI tones, which respect the mute.
- **Keys stay in the browser;** provider toggle with exportable per-run log; voice/test buttons.



- 30 trilingual technique cards, 5/day; missed topics repeat until correct.
- Streaks and crew accuracy; labeled sponsor slot appears only after answers.



- Centered column, all CTAs in-fold at 1440×900; camera grid reflows to two columns.
- Mobile-first, desktop-clean — same build, no separate site.

Judge's scorecard — how I would grade this today

Reading this entry as an Indie-track judge: it clears the pass/fail gate cleanly (fits **Build for Your World** in one sentence; uses Muse Spark + SAM 3.1 + voice transcription + Baseten Spark Open; stays clear of the "simple image analyzer" anti-category by shipping a full closed loop). Scored against the 0–10 rubric:

Criterion (weight)	Score today	Why	To reach 9.5–10 (what you must bring on Nov 4 / Nov 9)
Impact (30%)	**8.5 / 10**	Names the human (2 a.m. shift cleaner), the portability (record travels job to job), the anti-surveillance architecture (consent + read-aloud, faces never identified, read-only appeal, top-10-only board), and ships in EN/ES/PL with 44px glove targets. All 15 MVP screens and 10 company rules are in the build.	Show one real room cleaned and scored in the video, and name the Chicago pilot partner or target site count in the closing 15 seconds — deployed evidence is the difference between 8.5 and 9.5.
Meta AI Tooling (25%)	**9.0 / 10**	Genuinely beyond a wrapper: structured per-surface Spark scorecards with a 0.6 confidence floor (unverified is stated, never smoothed over), SAM 3.1 pixel overlays, voice transcription, a Baseten Spark Open toggle with an exportable per-run audit log, plus an on-device MediaPipe + classical-vision rung.	In the video, show *both* rungs back-to-back (on-device run, then a keyed Spark + SAM run with the overlays visible and the provider log exported). Without pre-entered keys a judge only sees Rung 1.
Demo (25%)	**8.5 / 10**	One-tap Judge Demo lands the scoring story in 10 seconds with zero pollution; the live site is up on GitHub Pages; the spectator board is built for stage projection.	The 3-minute submission video is the primary judging artifact — shoot the shot-list below on a real phone in a real breakroom/restroom, with the spectator board in frame.
Technical Depth (20%)	**9.0 / 10**	~11k lines of dependency-free vanilla JS, no build step, hermetic 6/6 self-test + persona suite, offline queue that re-scores on reconnect, 44px floors verified across 9 mobile screens + desktop, honest fallbacks everywhere.	Keep the disclosed-limitations list in the submission write-up — judges reward builders who know their own edges.

Weighted total	**8.73 / 10**	Strong finalist contender as-is; crosses **9.3+** once paired with the 3-minute real-room video and a pre-keyed live finals browser.

The 3-minute submission video — shot-by-shot

1. **0:00–0:20 · The world (Impact):** open on the phone in Spanish, in a real breakroom. Say the pitch line verbatim; tap to English; point at the always-visible streak/week/matchup strip and the consent read-aloud button.
2. **0:20–0:45 · The 10-second scoring story (Demo):** Instructions → tap *Judge Demo*; land on the scorecard; point at the three bars (50/25/25), the demo label, and the one-tap Dispute button ("the system scores, nobody edits").
3. **0:45–1:45 · Real clean, real models (Tooling):** select the registered site, scan the printed `CQ-QR` sticker, frame the sink with the guide, tap Start (with Hype set to Full so judges hear the clip chain), wipe the counter, match the angle with the ghost overlay, finish. Show the **Muse Spark** per-surface scorecard + **SAM 3.1** overlays + spoken voice note transcribed + provenance line.
4. **1:45–2:25 · Second provider + paperwork that writes itself (Tooling + Impact):** flip Settings to **Baseten Spark Open**, export the provider log JSON; open Records, export the geofenced inspection CSV, switch active squads in Pane 13, assign a 5-day pack in Pane 15.
5. **2:25–3:00 · Daily habit + stage close (Impact + Demo):** swipe two cards in the Daily Deck (show the brand-neutral answer, then the labeled `SPONSORED` suggestion); cut to the `?spectator=1` board on a monitor; close on the portable Passport copy toast.

Known limitations (disclosed, not hidden)

- **Site binding is honour-system at check-in:** runs must pick an admin-registered site (and carry QR provenance when scanned) to be prize-eligible, which closes the MVP rule; cryptographic per-site QR tokens are a post-pilot hardening item.
- **Headless ≠ device:** MediaPipe needs real WebGL; CI/headless runs show the honest fallback path. Judge on a real phone.
- **Cloud needs keys:** without operator keys judges see the on-device path only — which is itself a complete game, with the offline note shown.
- **Residual detector advisory:** one low 11px-label note in dense game chrome; body copy is 12–14px+.

Appendix — verification

- Layout audit: 9/9 mobile screens at 360×740 (zero sub-44px layout targets, zero h-scroll, primary CTAs in-fold) + desktop 1440×900 pass (zero sub-44, zero h-scroll).
- Independent judge + advisor reviews run against the operating standards; all blockers fixed and re-verified (forced-unmute removed and tones mute-gated; live zone spoilers neutralized; judge-demo cloud egress blocked; demo/sample runs excluded from ranking, personal best, and sponsor counts; offline queue re-scores on reconnect).
- `node --check` clean; detector 8 → 1 low advisory; no secrets in repo; banned-token grep clean.
- Self-test 6/6 PASS; persona suite 9/11 — both fails (*dana* record check, *fold* CTA check) reproduce identically on the pre-change tree: the harness runs headless at 800×600, below the 360×740 supported floor, so they are pre-existing viewport artifacts, not regressions. All 9 supported-viewport mobile audits and the desktop pass are green.
- Report screenshots are the app itself (390×844 mobile, 1440×900 desktop), captured 2026-10-31; leaderboard/spectator demo rows disclosed.